



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



Audience-dependent XAI in finance

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MSCA DIGITAL

Funded under the Marie Skłodowska-Curie Actions (MSCA), the network brings together academic institutions, financial companies, technology providers and policy organizations to jointly train the next generation of researchers in digital finance

Key thematic areas: **The European Financial Digital Space, AI-driven analysis of financial markets, Explainable and fair AI in finance, Blockchain infrastructure and innovation, Sustainability and green finance**

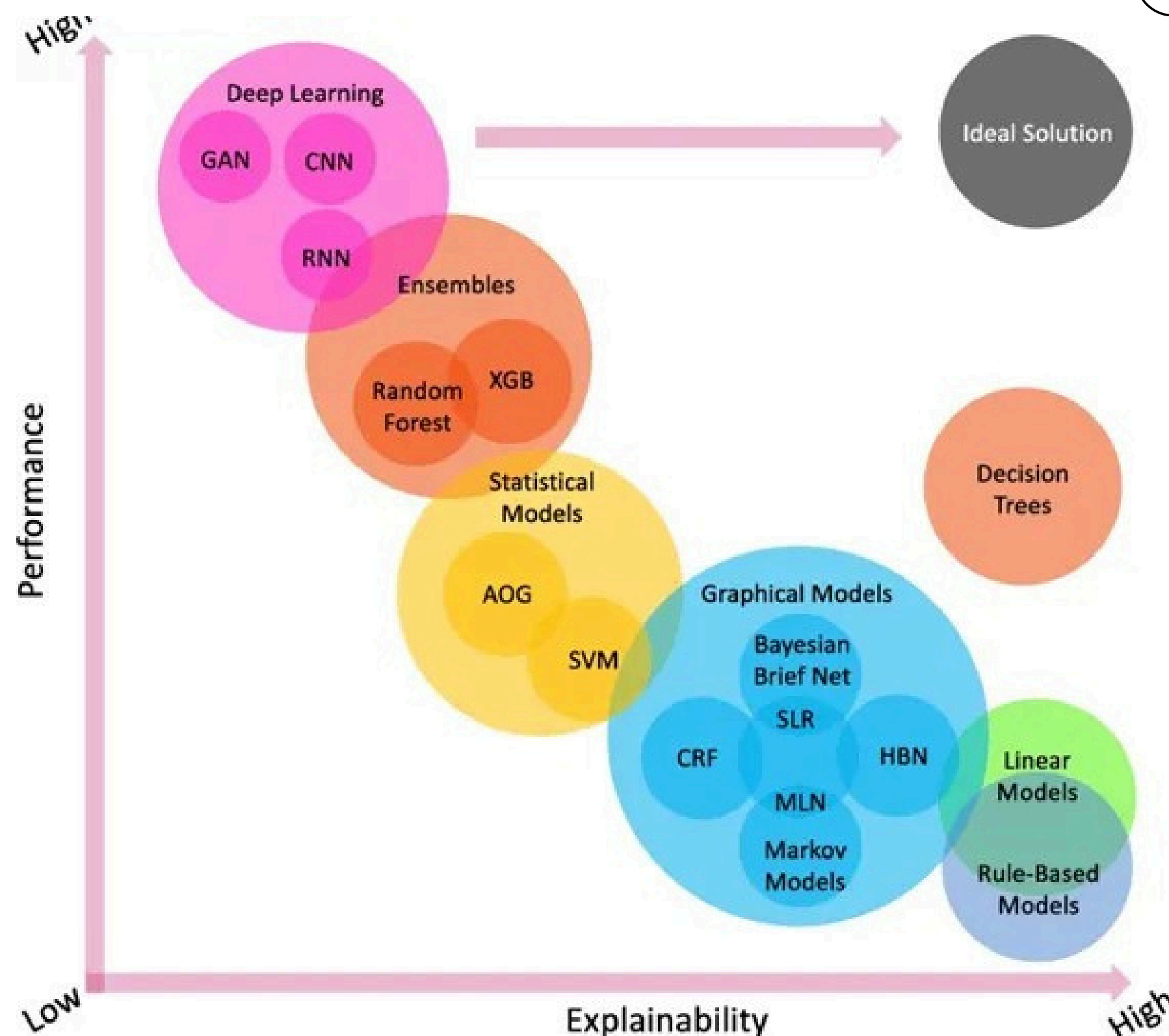


Some XAI Basics

What are blackbox models?

MODELS WHOSE INTERNAL LOGIC IS NOT DIRECTLY INTERPRETABLE BY HUMANS

- "White-box": linear regression, decision trees vs "black-box" neural networks or ensemble methods
- Literature in finance (see Zetzsche et al., 2020; Kumar, 2017; Cernevicene et al., 2024; Ogunniyi, et al., 2025)



Why choose black-boxes?

"THE CURSE & GIFT OF DEPTH"



Optimization

Models learn by minimizing the loss through gradient descent.



Non-linearity

Activation functions (e.g., ReLU, sigmoid) allow complex decision boundaries.



Overparameterization and emergent structure

Enables pattern discovery but hides reasoning

Deep layers encode abstract manifolds = statistically efficient, but cognitively opaque

Explainable AI

METHODS THAT MAKE BEHAVIOR, DECISIONS,
AND INTERNAL LOGIC UNDERSTANDABLE
TO HUMANS



Scope

Global - understand the overall behavior of the model

e.g., feature importance, partial dependence plots (PDP)

Local - explain a single prediction
e.g., LIME, SHAP, counterfactuals

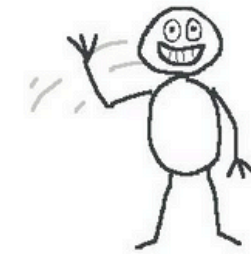


Dependence

Model-specific - works with only certain algorithms (e.g., coefficients in linear / logistic regression, tree visualizations)

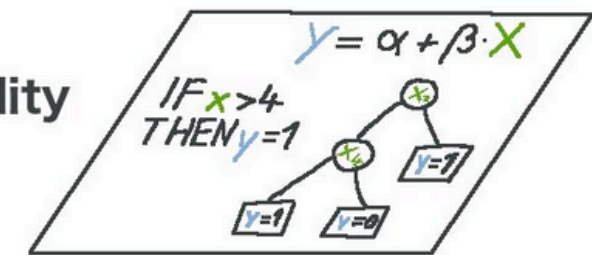
Model-agnostic - works with any model by proving input-output behavior (e.g., SHAP, LIME)

Humans



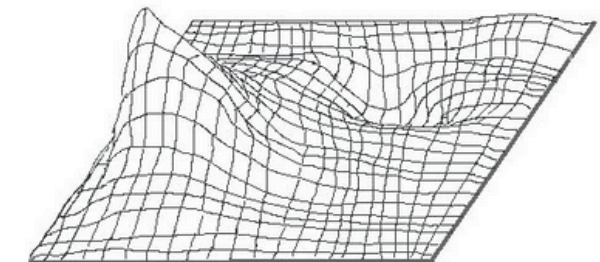
↑ inform

Interpretability
Methods



↑ extract

Black Box
Model



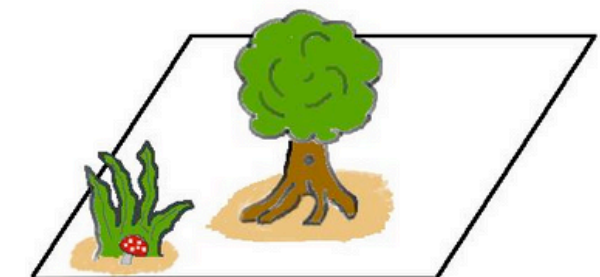
↑ learn

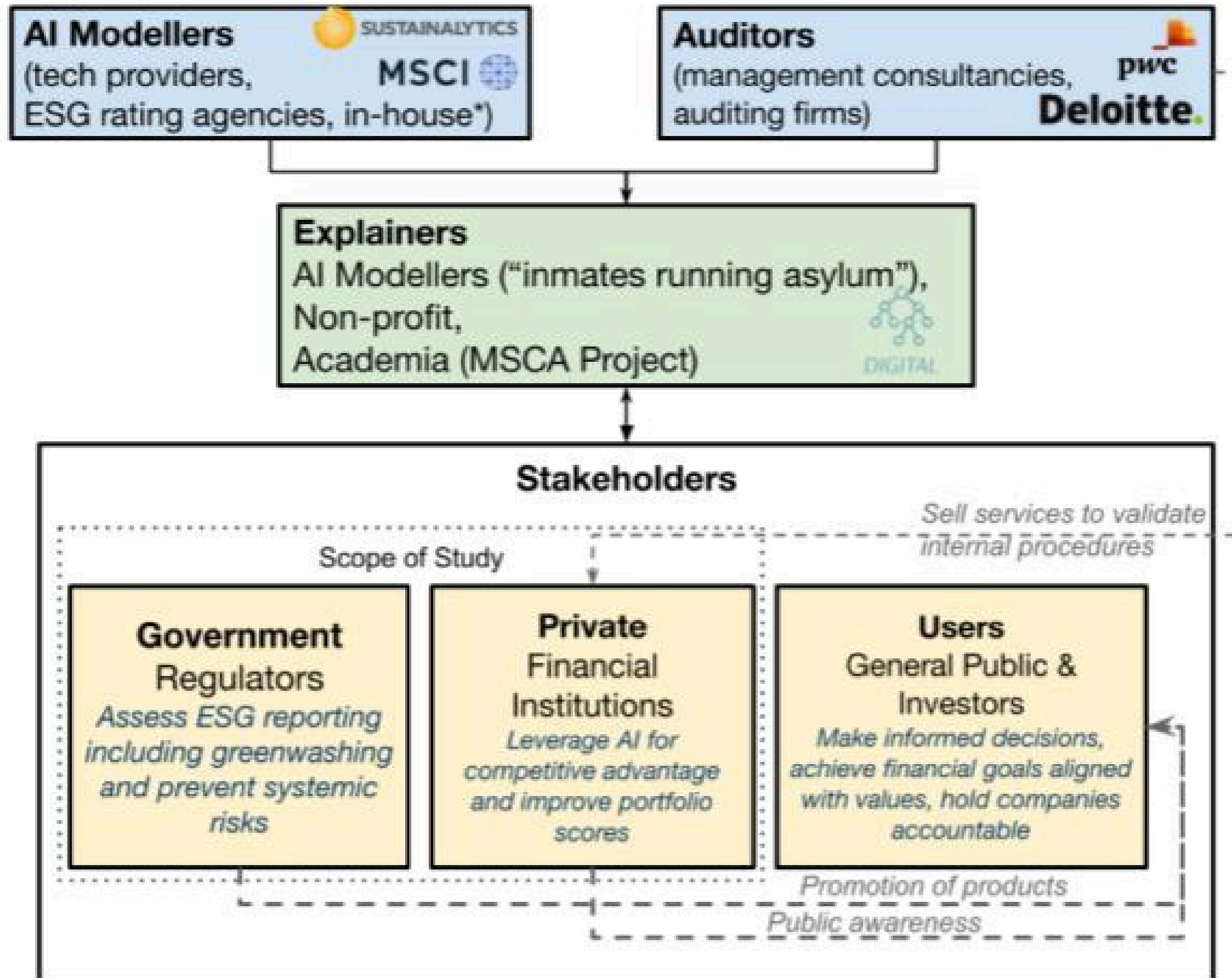
Data

X_1	X_2	X_3	...	X_n
1	2	0	...	5
2	1	0	...	4
3	0	1	...	3
...	...	...	...	...
n	n	n	...	n

↑ capture

World





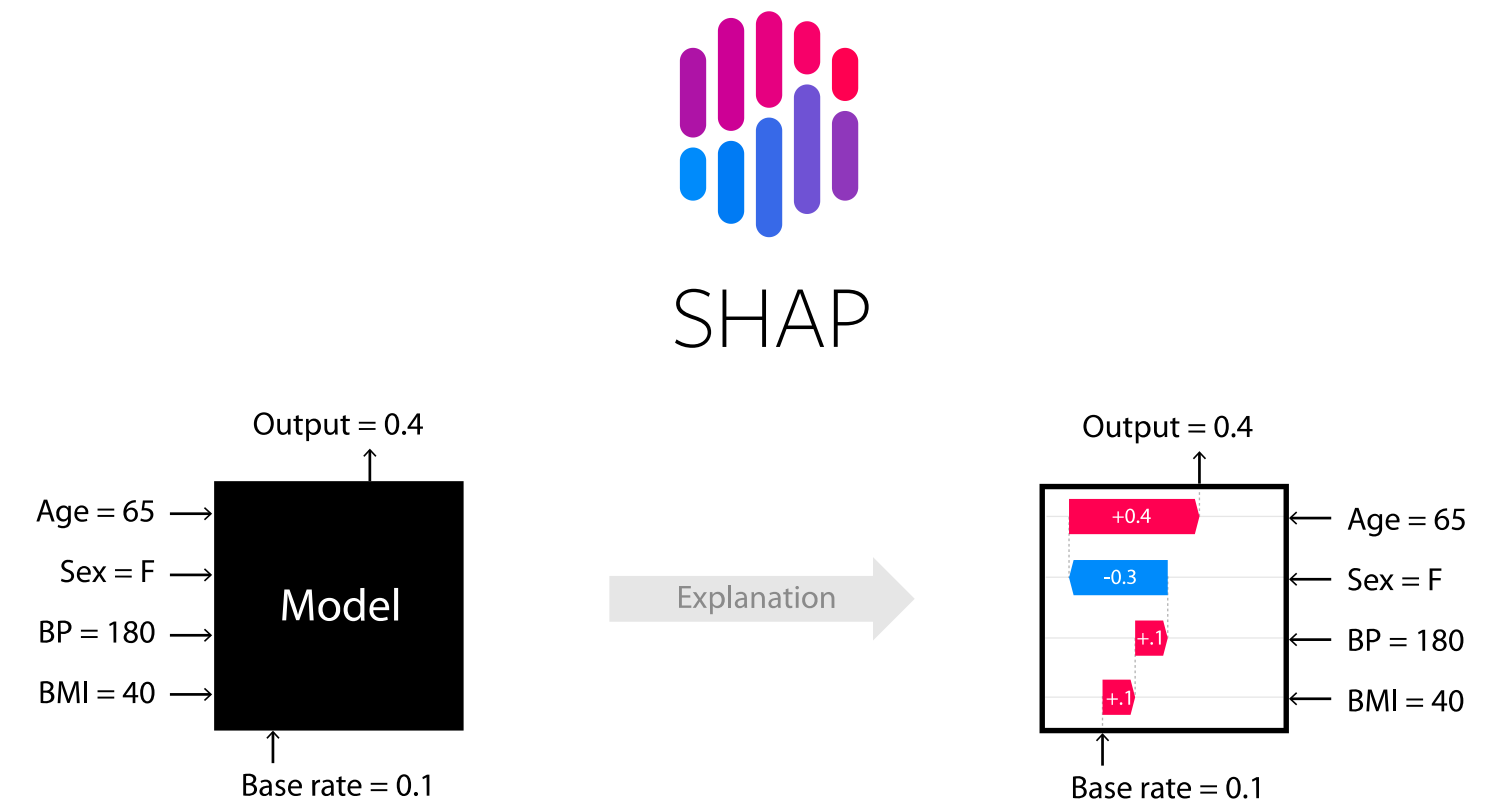
Research Problem

WHAT I WANT TO SOLVE

XAI methods in finance are focused on technical users (SHAP, LIME) like "inmates running the asylum" (Miller et al., 2017)

Incomprehensible AI leads to low AI trust and adoption (Bernardo & Seva, 2023)
Overtrust in incapable AI (Ehsan et al., 2021)

Literature suggests audience-dependent explanations (Misheva & Osterrieder, 2023; Schmude et al., 2024)



"[On shapley values] it is not
intuitive enough for a user to
explain it." — B2B Sales Officer, AI Vendor (Europe)

“The business would not understand it. They were so used to
the regression coefficients... if you increase this variable,
therefore the value will increase.”

— Data scientist, Bank (Asia)

Roles in Interpretable AI

Molnar (2025) and author



(Affected)
User

e.g.
customers/
public



Operator

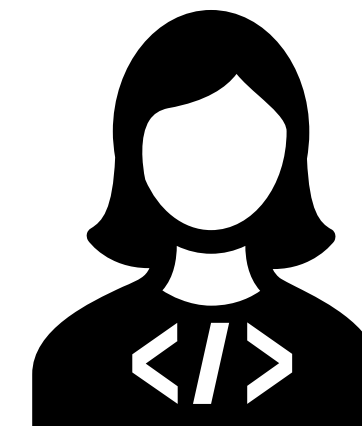
aka
business
users
(marketing,
compliance,
etc.)

Executor

makes
decisions
based on
AI

Manager

responsible
for the
project



Developer

creator

Validator

checks
model's
outputs



Regulator



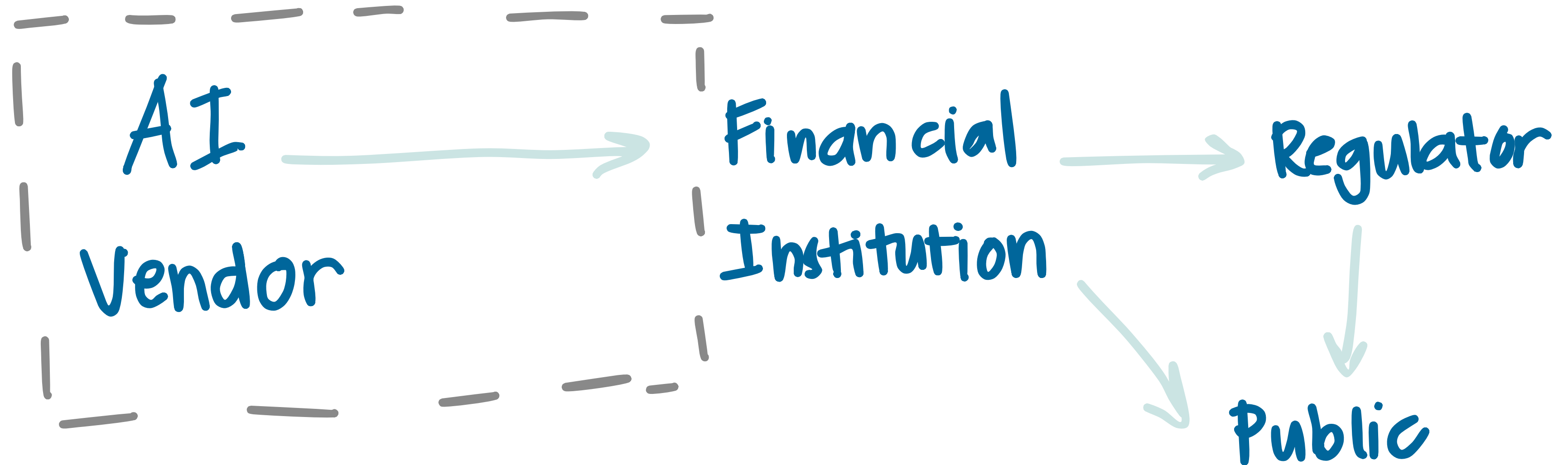
Conduct

consumer
protection,
fair trade

Prudential

systemic
financial
risk (ECB)

In-house vs proprietary AI tools



→ Direction of
XAI accountability

Preliminary Findings

Finance Professionals

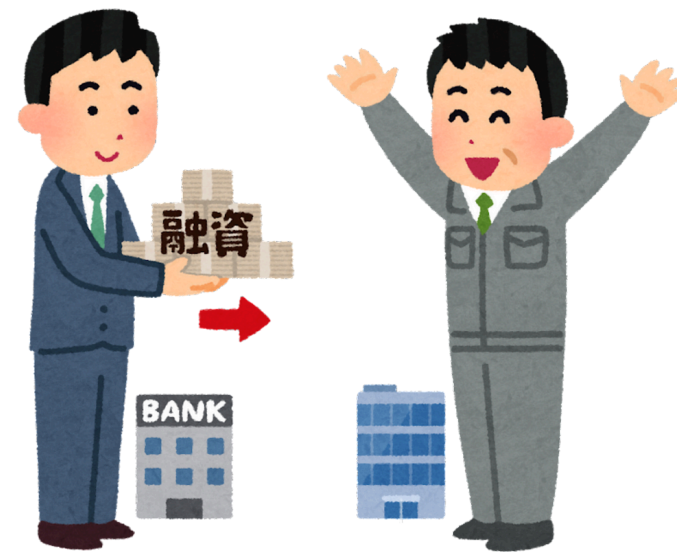
State of black box models

Exploratory data analysis

Hindered by regulators in high-risk decisions

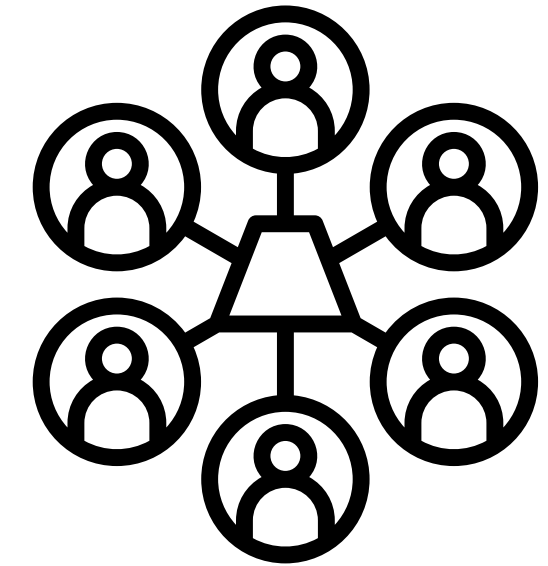
“We use black box models... but in production, we only use logistic regression based models... because [the ECB] won't allow it.”

Added more variables for credit scoring,
otherwise not in mainstream financial theory





Emerging themes



Human-in-the loop

“It was one of the reasons that they said we had to remodel is because **they didn't want it completely by the algorithm.**”

“So it will give a trigger, and then a human will determine what to do with it.”

&

Embedded roles

“We say to the client OK....you need to properly explain it before you give this to the regulator.”

“The explainability and all those things are not for the end user, but for the regulator mostly.”

We keep track of this kind of testing... factualness score when we change the question and answer... also for the regulator.

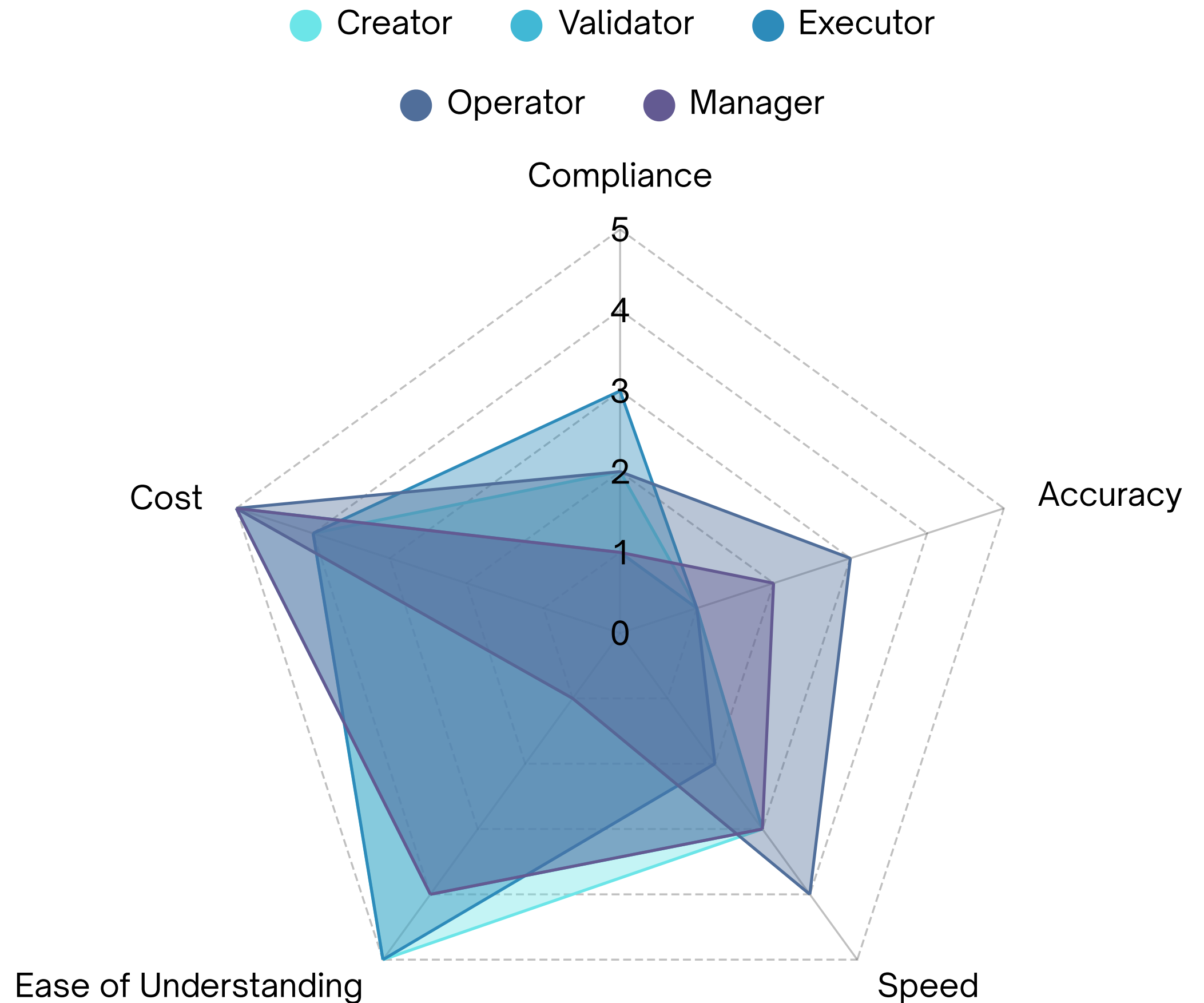
Tradeoffs in Explainability

What research say		Borda Count (Weighted points for 10 participants)
Priority with the GDPR (Misheva et al., 2021; David, 2021), professional standards (du Preez et al., 2024), and EU AI Act.	Compliance	46
- Concerns about consequences in audit trail (Muller et al., 2022), high error rates (Nakashima et al., 2022) - Recommended need for human oversight to address (Zetzche, n.d.; Bertrand, 2024)	Accuracy	40
- Challenge in real-time scaling (Deo and Sontakke, 2021; Boggavarapu, 2025) - Time spent in understanding information (Gerlins and Constantiou, 2022) issue in fast-paced financial environments	Speed	29
- Information overload and processing difficulties (Gerlings and Constantiou, 2022) - Varying AI literacy levels among managers (Sabharwal et al., 2024) and actuaries (du Preez et al., 2024)	Ease of understanding	21
Cost and economic efficacy as barrier to adoption (Adamyk, 2024), affecting profitability (Kelley, 2022)	Cost	15

Trade off by role

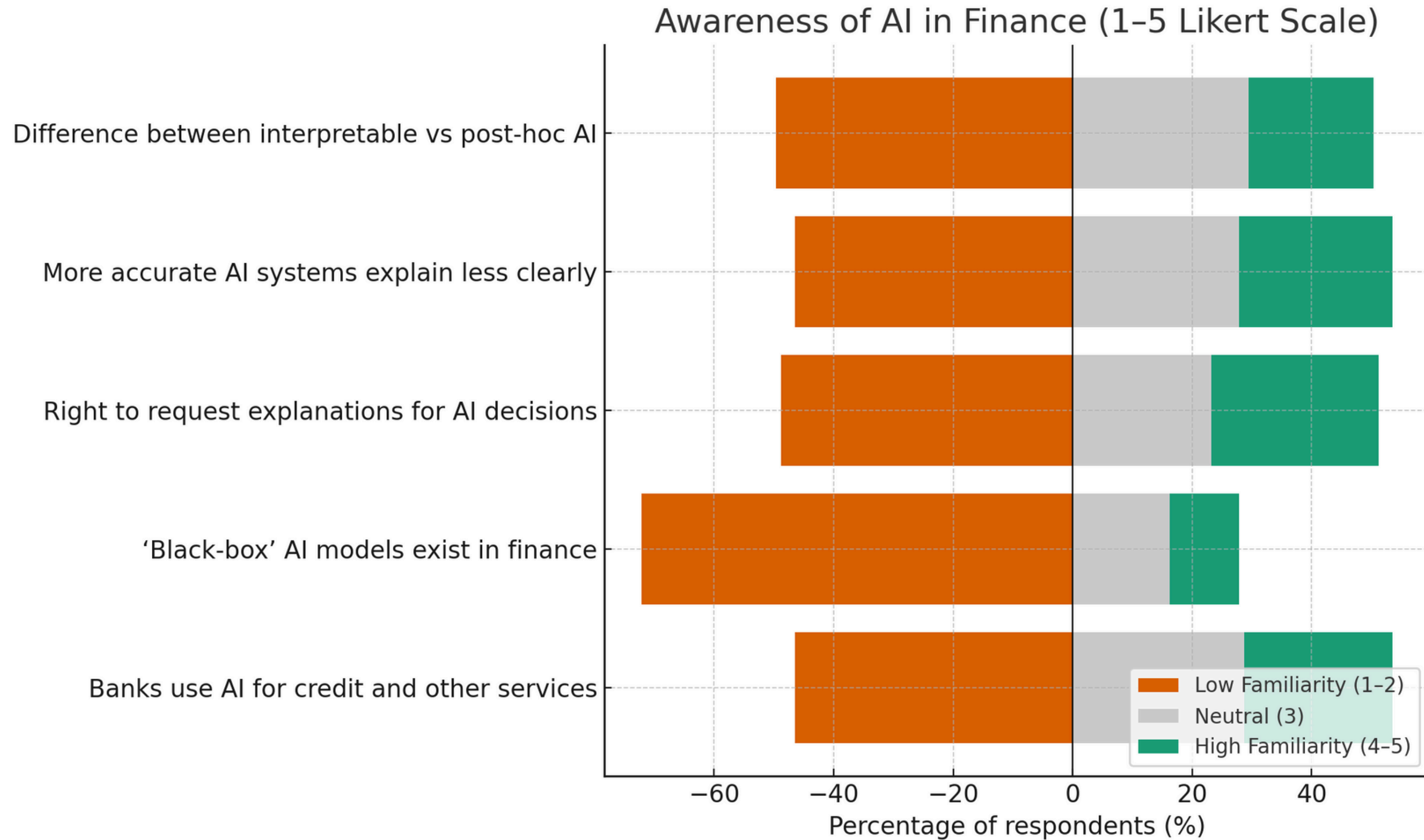
Borda Count scores were normalized into rank positions (1 = highest priority, 5 = lowest) for each feature within each role.

Roles rank **Accuracy** and **Compliance** highest, with Creators and Validators prioritizing model quality, and Managers focusing on **regulatory** fit. Operators value **Ease of Understanding**, while **Cost** ranks lowest across all roles.



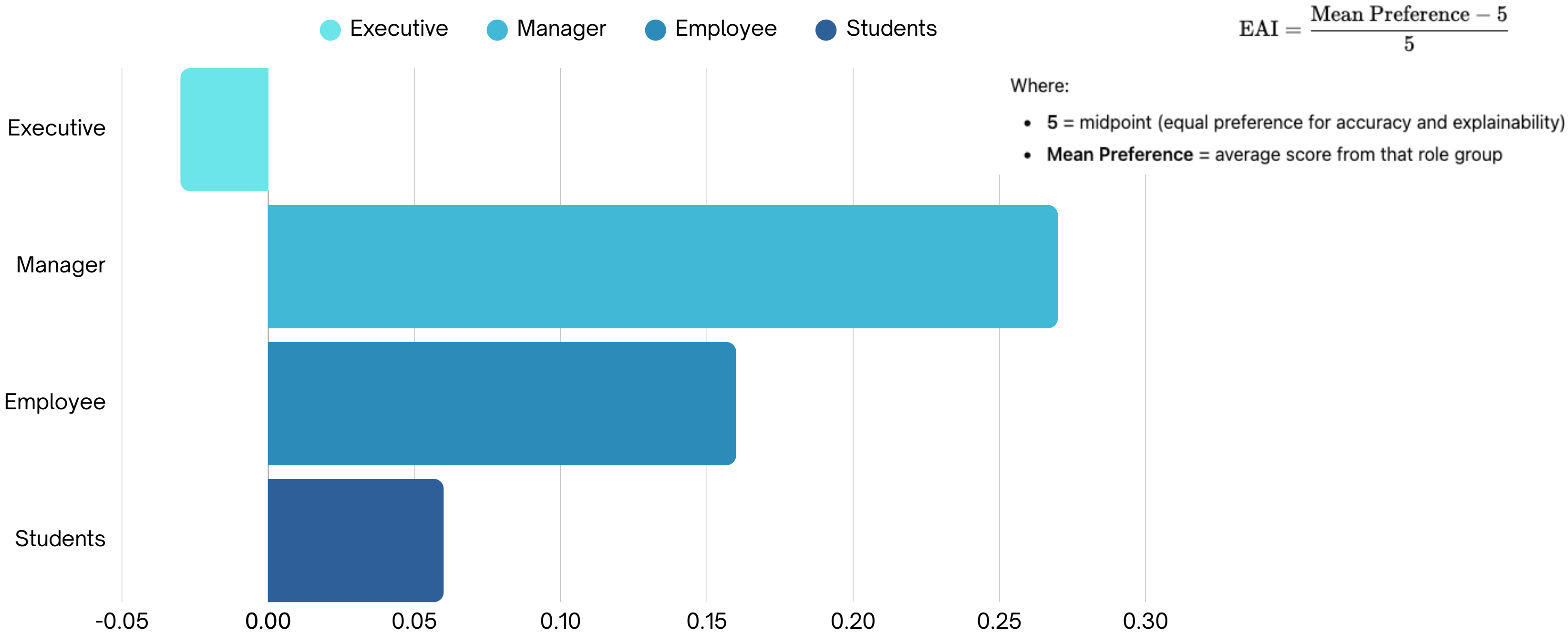
Preliminary Findings

Public / App Users

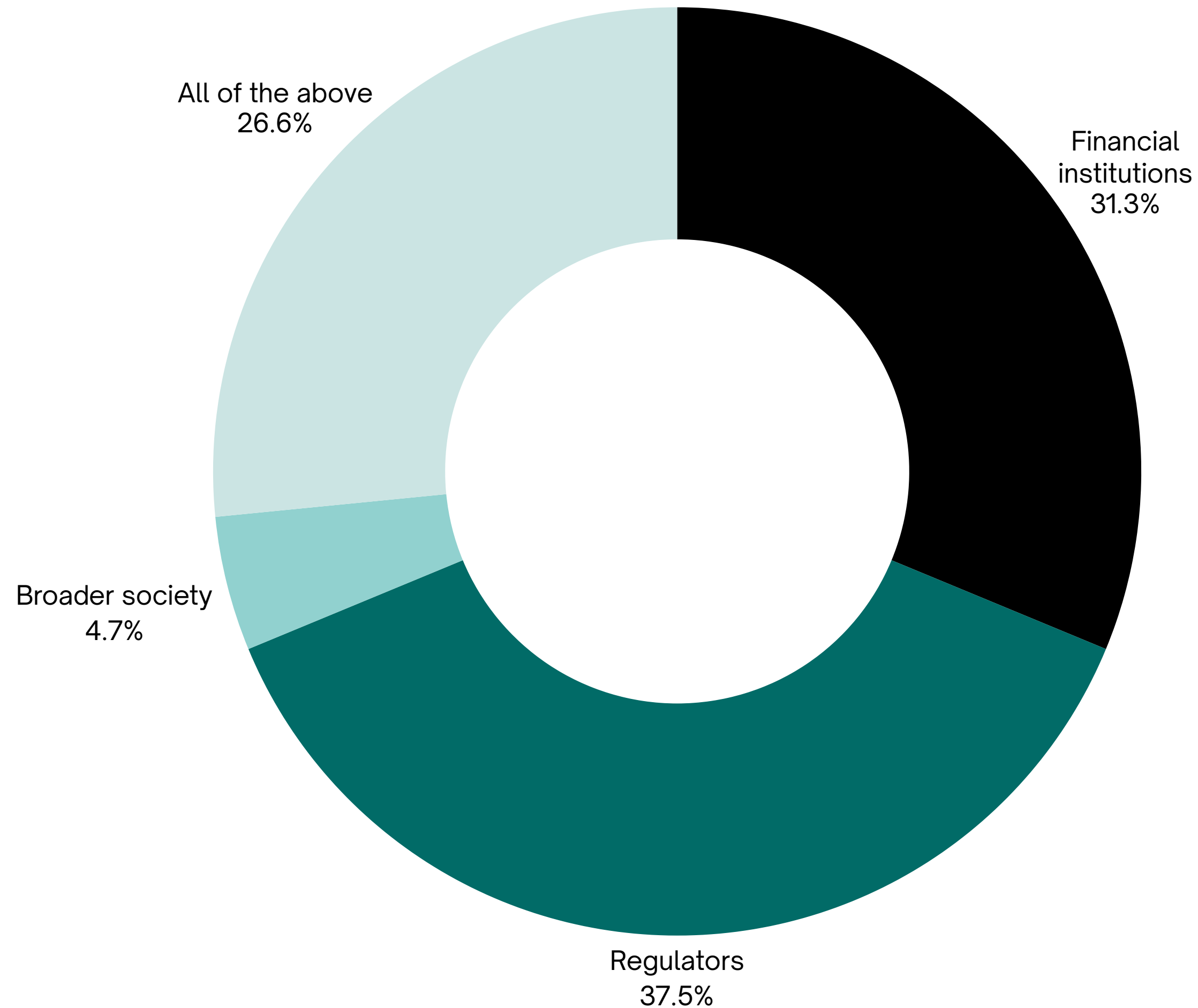


Executives value accuracy over explainability.

Participants rated their ideal AI system from 0 (accuracy-focused) to 10 (explainability-focused).

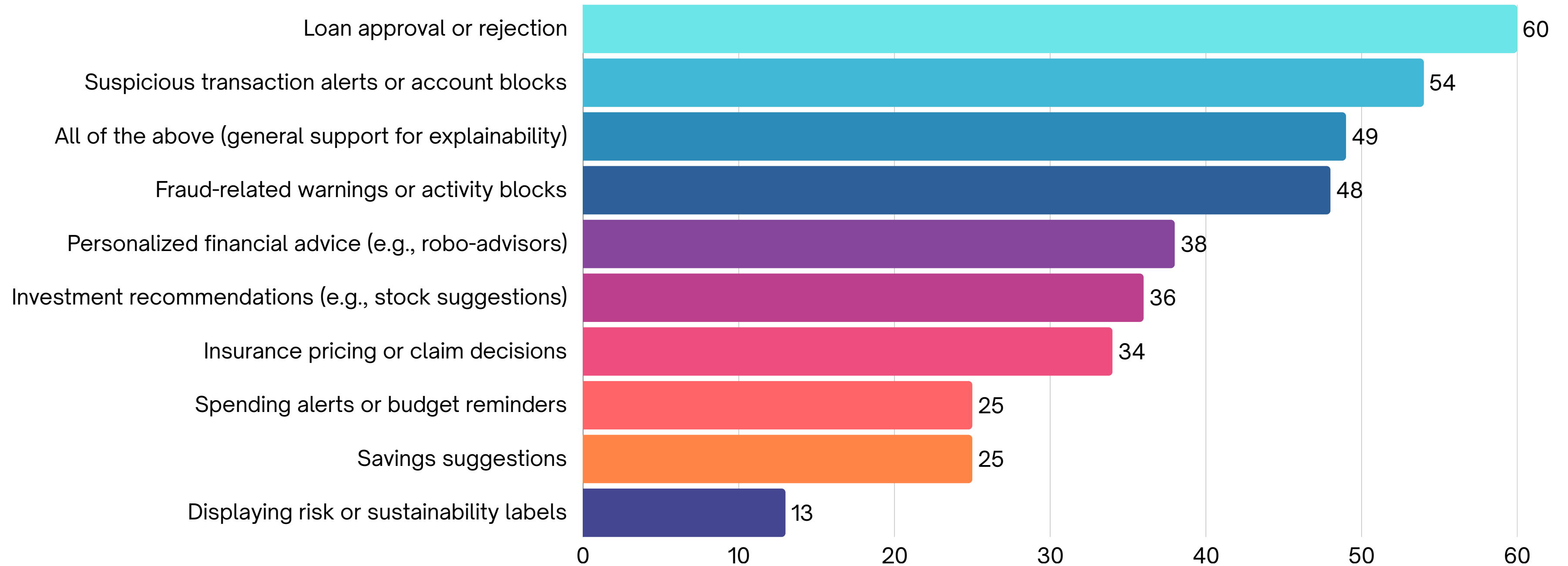


Who should be responsible for ensuring AI transparency in financial apps?



Which financial transactions should have mandatory explanations?

The strongest demand for explainability appears in high-impact, decision-critical areas, especially credit scoring (loan approvals) and fraud management.



Next steps

October 31

Feedback on pilot surveys



November 3

Launch of financial professionals survey

November 15

Launch of general public survey

Early 2026

Focus group discussion with central bankers, regulators, government

2026-2027

Role-based XAI recommender system

Thank you!